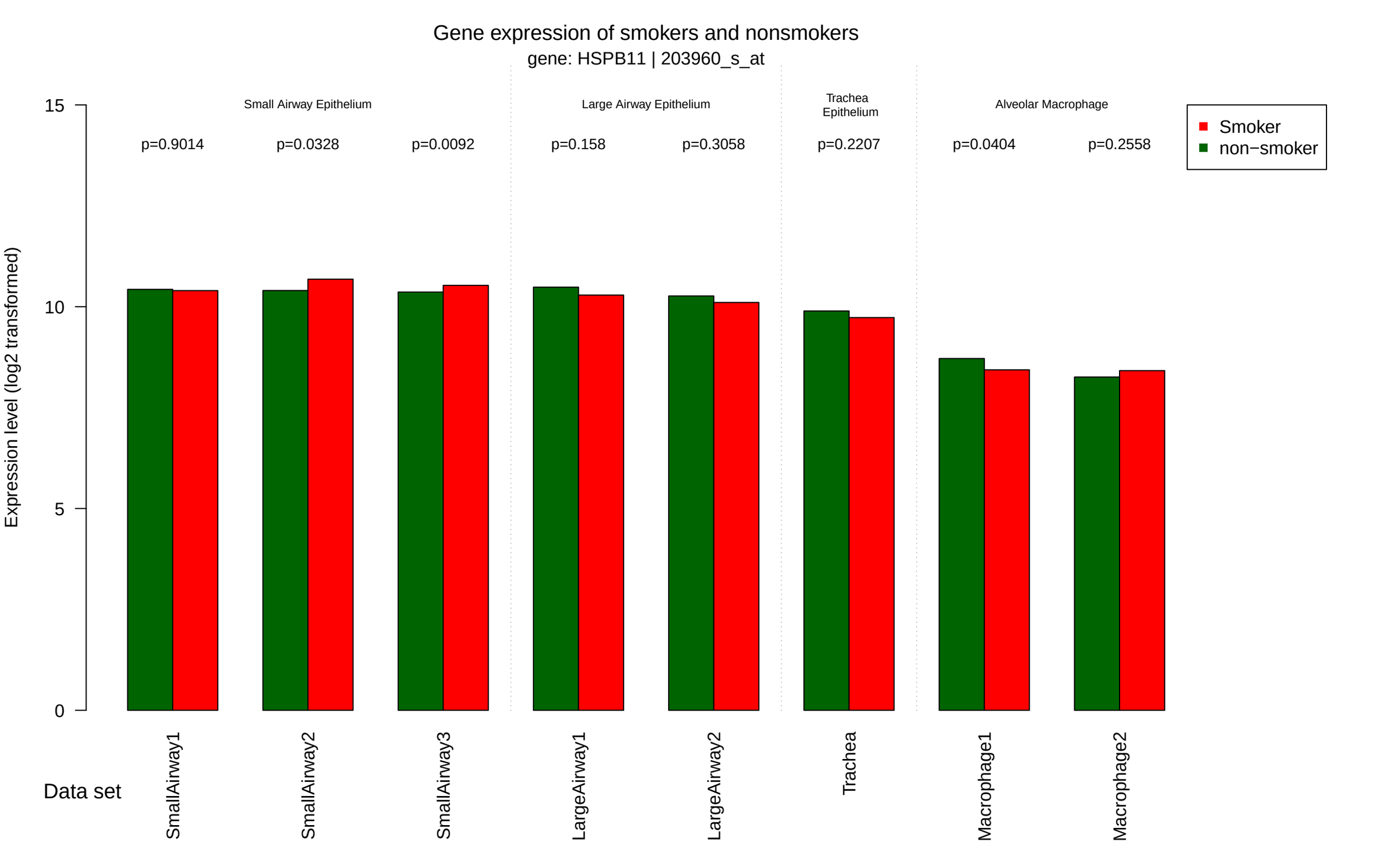
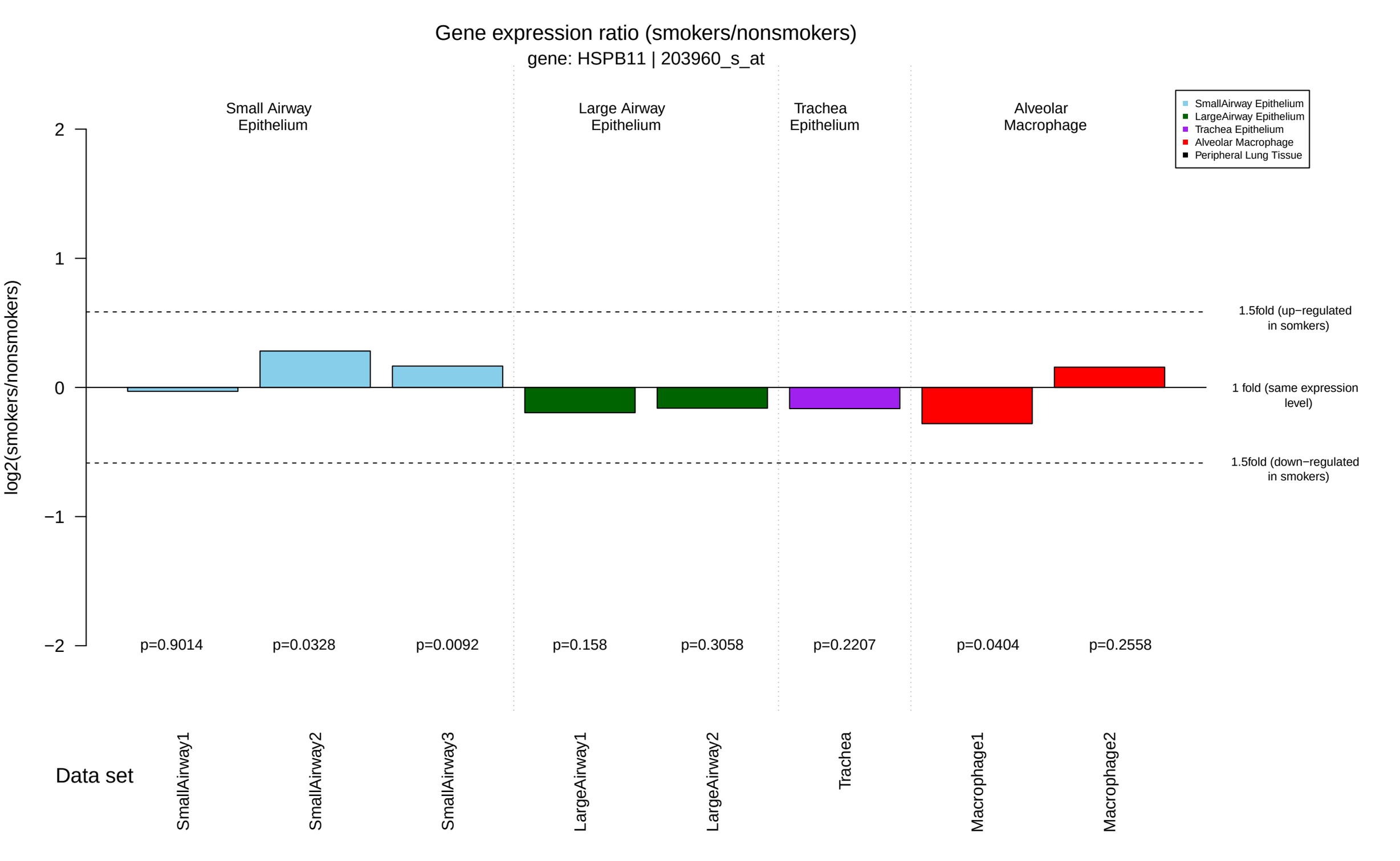
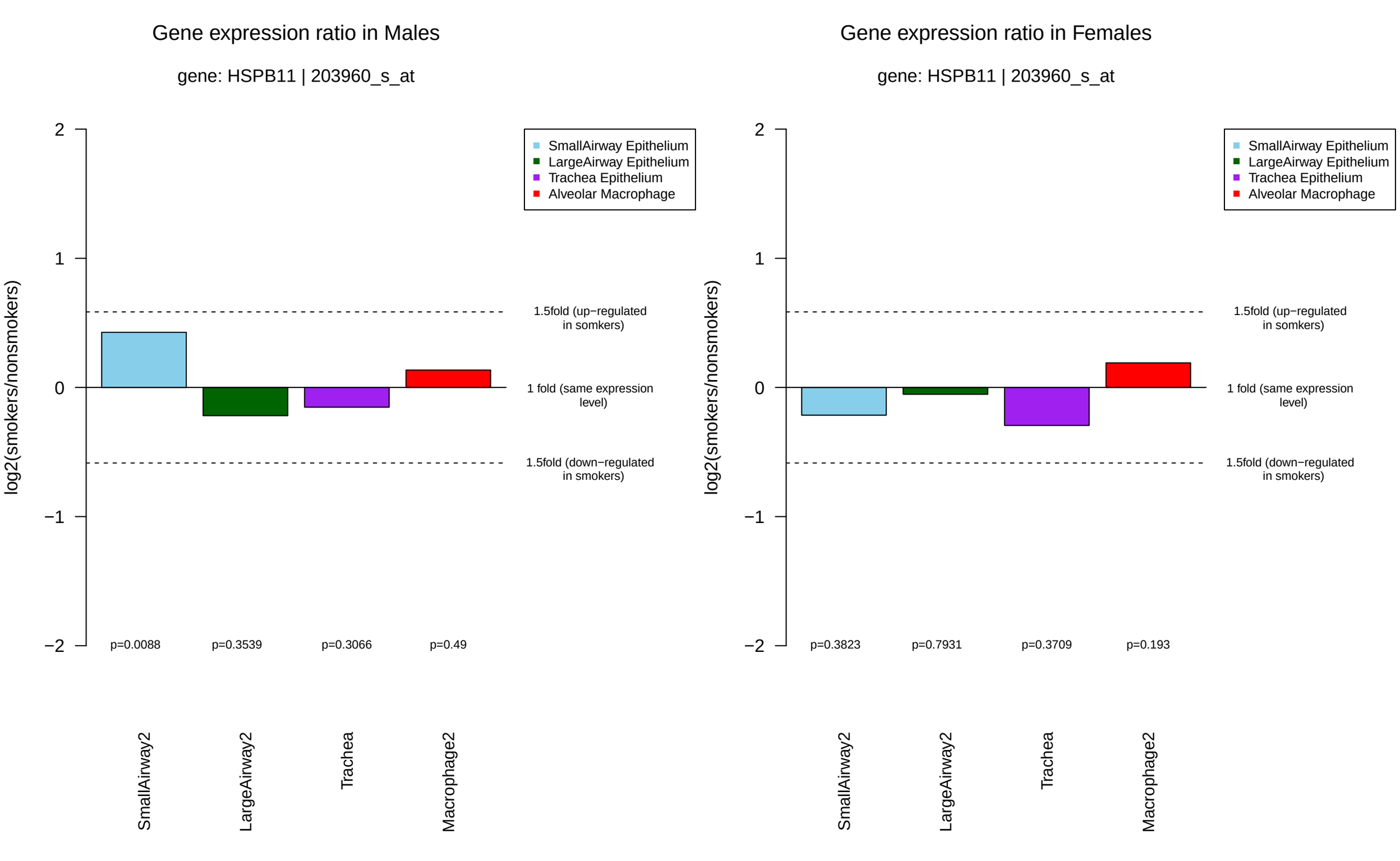
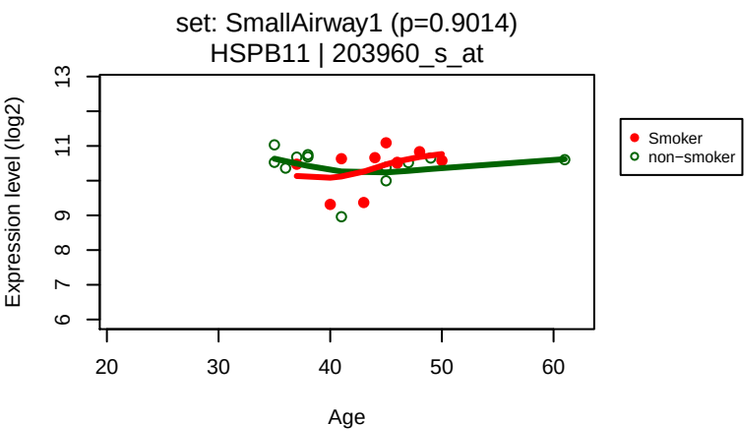
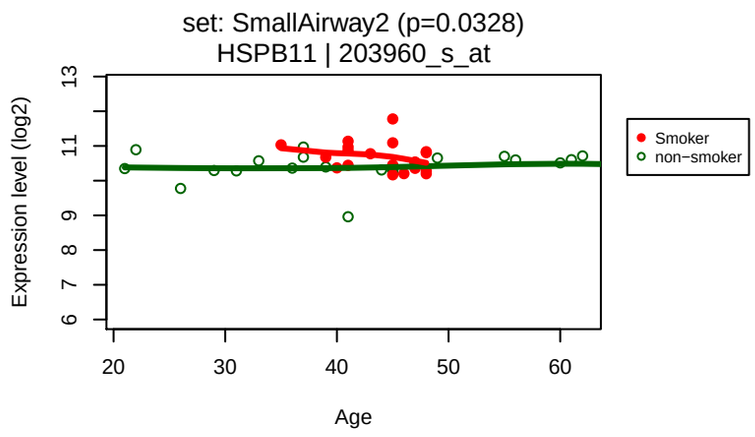
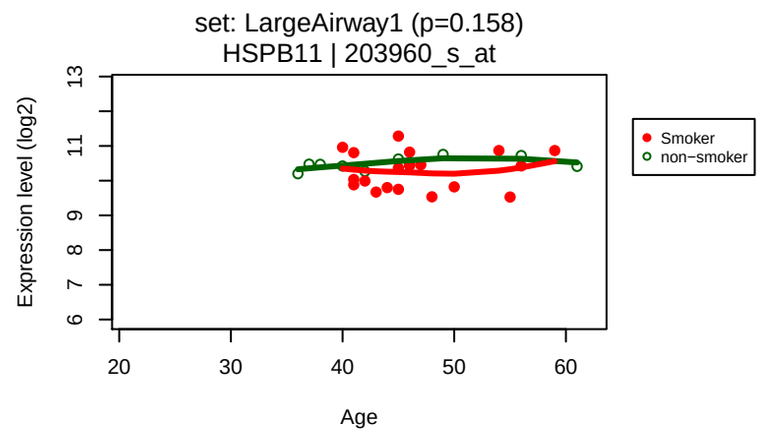

Gene expression of smokers and nonsmokers
gene: HSPB11 | 203960_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9014
p=0.0328
p=0.0092
p=0.158
p=0.3058
p=0.2207
p=0.0404
p=0.2558
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

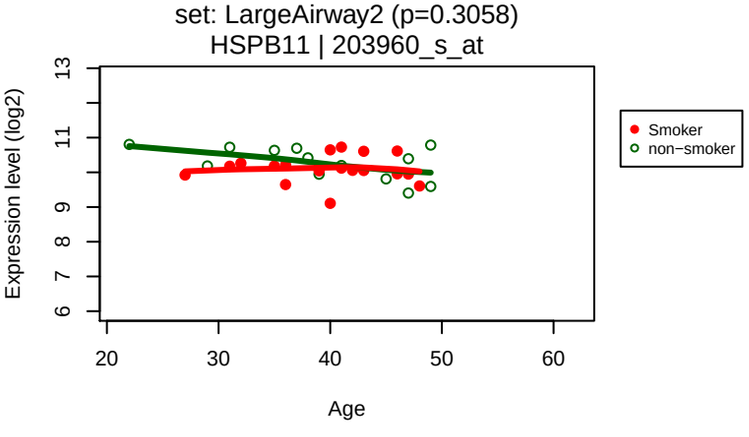

Gene expression ratio (smokers/nonsmokers)
gene: HSPB11 | 203960_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9014
p=0.0328
p=0.0092
p=0.158
p=0.3058
p=0.2207
p=0.0404
p=0.2558
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

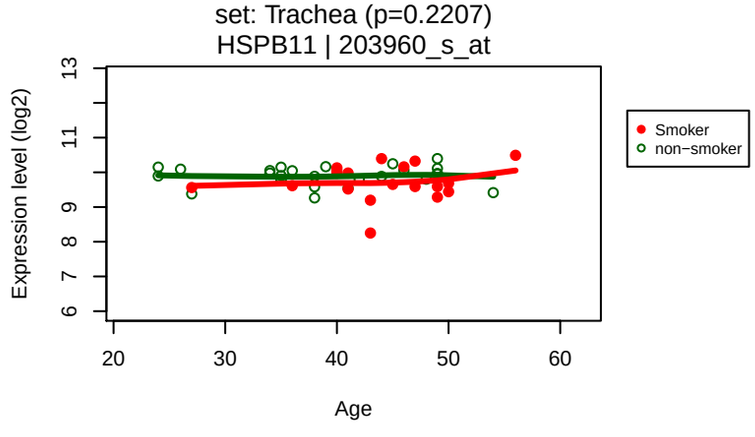

Gene expression ratio in Males
gene: HSPB11 | 203960_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0088
p=0.3539
p=0.3066
p=0.49
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPB11 | 203960_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3823
p=0.7931
p=0.3709
p=0.193
SmallAirway2
LargeAirway2
Trachea
Macrophage2

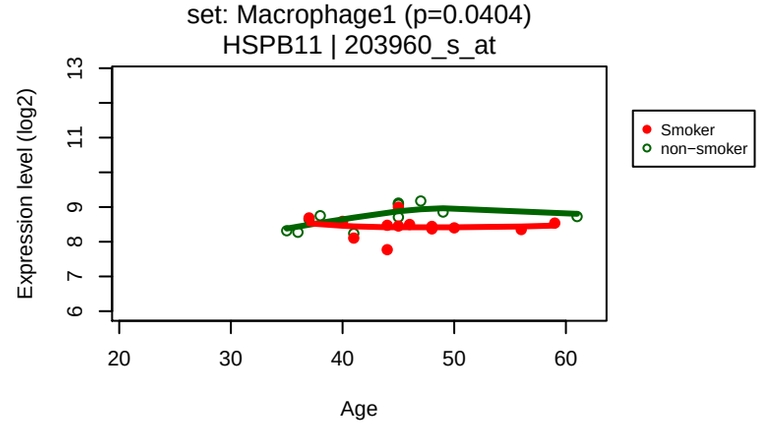

set: SmallAirway1 (p=0.9014)
HSPB11 | 203960_s_at
Expression level (log2)
Age
Smoker
non−smoker

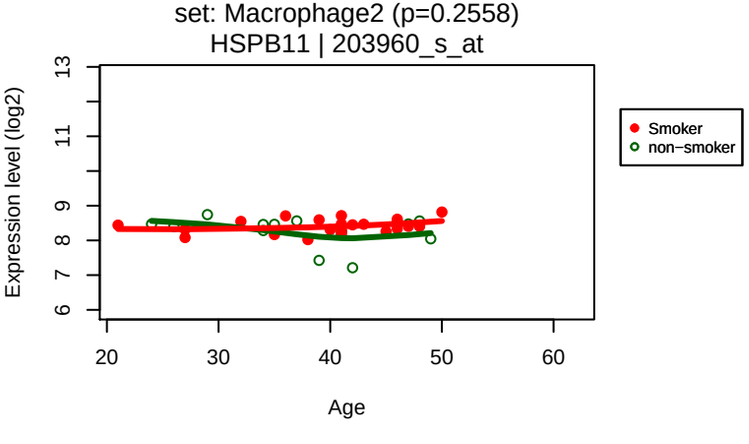

set: SmallAirway2 (p=0.0328)
HSPB11 | 203960_s_at
Expression level (log2)
Age
Smoker
non−smoker

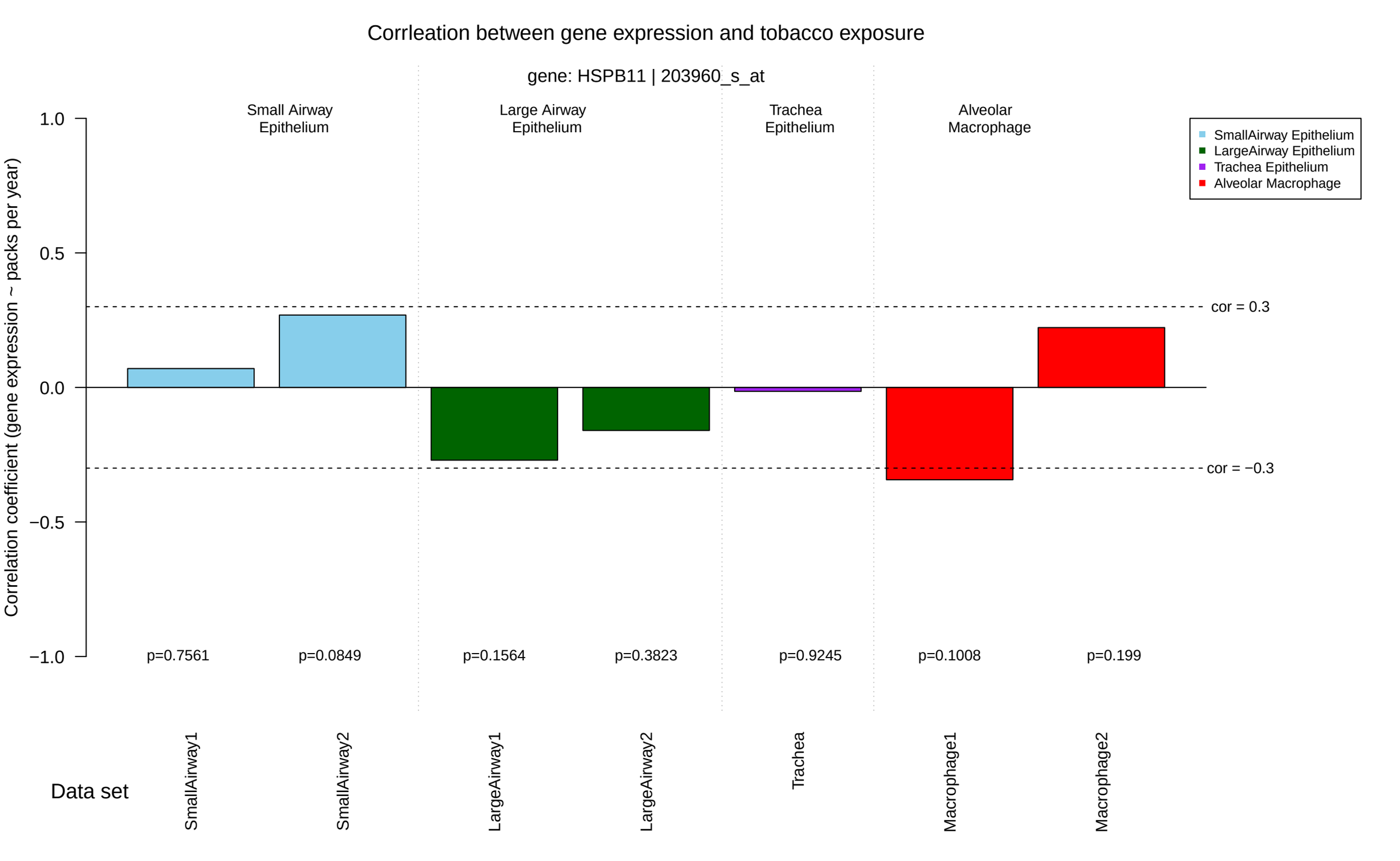

set: LargeAirway1 (p=0.158)
HSPB11 | 203960_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3058)
HSPB11 | 203960_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2207)
HSPB11 | 203960_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0404)
HSPB11 | 203960_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2558)
HSPB11 | 203960_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPB11 | 203960_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7561
p=0.0849
p=0.1564
p=0.3823
p=0.9245
p=0.1008
p=0.199
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2